Reanalysis of Scheibehenne, Jamil, & Wagenmakers (2016)

Hotel towel reuse

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Overview

- Computes joint BFs for the set of experiments reported in from Scheibehenne et al. (2016).
- Source paper: Scheibehenne, B., Jamil, T., & Wagenmakers, E.-J. (2016). Bayesian Evidence Synthesis Can Reconcile Seemingly Inconsistent Results: The Case of Hotel Towel Reuse. Psychological Science, 27(7), 1043-1046.
- Each experiment is analyzed as one 2x2 contingency table of counts:
  - rows: condition (control, social norm)
  - columns: towel reuse outcome (yes, no)
- For each study, evidence is computed for:
  - predicted hypothesis: social norm > control (higher reuse probability)
  - unconstrained hypothesis
  - complement: social norm < control

```r
library(BayesFactor)
library(dplyr)
library(here)

source(here("functions/compute_joint_complements.r"))

# helper functions for formatting numbers
fmt2 <- function(x) sprintf("%.2f", x)
fmt4 <- function(x) sprintf("%.4f", x)
fmtE <- function(x) sprintf("%.2e", x)
```

Data

```r
rawData <- read.table(
  here("scheibehenne_original/towelData.csv"),
  header = TRUE,
  sep = ";",
  stringsAsFactors = FALSE
)

# n experiments, each represented by 4 rows (from 2x2 contingency table)
n <- nrow(rawData) / 4

# Add pooled "Total" 2x2 table as in the original script
bigTable <- array(rawData$Count, dim = c(2, 2, n))
total2x2 <- apply(bigTable, 2, rowSums)

totalFrame <- data.frame(
  Source = rep("Total", 4),
  AuthorName = rep("Total", 4),
  Experiment = rep(NA, 4),
  Year = rep(NA, 4),
  Group = c("Control", "Social Norm", "Control", "Social Norm"),
  Towel.Reuse = c("Yes", "Yes", "No", "No"),
  Count = c(total2x2[1, 1], total2x2[2, 1], total2x2[1, 2], total2x2[2, 2])
)

rawData <- rbind(rawData, totalFrame)
```

Study-specific Bayes factors

```r
# Names of experiments
exList <- unique(rawData$Source)

# Initialize vectors to store study-specific Bayes factors
bf_1u_vec <- rep(NA_real_, length(exList))
bf_cu_vec <- rep(NA_real_, length(exList))
bf_10_vec <- rep(NA_real_, length(exList))
```

```r
bf_1c_vec <- rep(NA_real_, length(exList))

# Set seed for reproducibility
set.seed(45)

# Compute BFs for each experiment
for (i in seq_along(exList)) {

  # Extract the 2x2 contingency table for the current experiment:
  # rows = condition (control, social norm), columns = towel reuse (yes, no)
  thisEx <- rawData[rawData$Source == exList[i], ]
  table2x2 <- cbind(yes = thisEx[1:2, "Count"], no = thisEx[3:4, "Count"])
  rownames(table2x2) <- c("control", "socialNorm")

  # Fit the unconstrained model H_u against the null model H_0:
  # - sampleType = "indepMulti", fixedMargin = "rows": the number of guests per
  #   condition is fixed, so each condition is an independent binomial with its
  #   own reuse proportion (theta_c for control, theta_e for social norm)
  # - priorConcentration = 1: independent uniform Beta(1, 1) priors on theta_c
  #   and theta_e (as in Scheibehenne et al., 2016, p. 1043)
  # - H_0 states that both conditions share the same proportion
  X <- contingencyTableBF(
    table2x2,
    sampleType = "indepMulti",
    fixedMargin = "rows",
    priorConcentration = 1
  )

  # Unused draws, kept from the original script: they consume random numbers,
  # so removing them would change the posterior samples below (and thus the
  # reported Bayes factors)
  post_alt  <- posterior(X, iterations = 10000)
  post_null <- posterior(1 / X, iterations = 10000)

  # BF_u0: evidence for H_u against H_0 (computed analytically by BayesFactor;
  # X stores it on the log scale)
  bf_u0 <- exp(as.numeric(X@bayesFactor$bf))

  # Draw 10,000 posterior samples of the two proportions under H_u
  # (these follow Beta(1 + yes, 1 + no) distributions in each condition)
  post.result <- BayesFactor::posterior(X, iterations = 10000)
  idx <- grep("omega", colnames(post.result))
  theta <- as.data.frame(post.result[, idx])
  # column 1: theta_c, proportion of towel reuse in the control condition
  # column 2: theta_e, proportion of towel reuse in the social-norm condition

  # Fit of H_1 (theta_e > theta_c) and of the complement H_c (theta_e < theta_c):
  # the proportions of posterior samples in agreement with each hypothesis
  prop.consistent <- mean(theta[, 1] < theta[, 2])
  prop.complement <- mean(theta[, 1] > theta[, 2])

  # BF_1u = fit / complexity; the complexity is P(theta_e > theta_c) under the
  # uniform priors, which is 0.5 because the prior is symmetric
```

```r
  bf_1u <- prop.consistent / 0.5

  # BF_10 by transitivity: BF_10 = BF_u0 * BF_1u
  bf_10 <- bf_u0 * bf_1u

  # BF_cu for the complement, with complexity 1 - 0.5 = 0.5;
  # via BF_c0 = BF_u0 * BF_cu, so BF_cu = BF_c0 / BF_u0 = prop.complement / 0.5
  bf_c0 <- bf_u0 * prop.complement / 0.5
  bf_cu <- bf_c0 / bf_u0

  # BF_1c by transitivity: BF_1c = BF_1u / BF_cu
  bf_1c <- bf_1u / bf_cu

  # Store the study-specific BFs
  bf_1u_vec[i] <- bf_1u
  bf_cu_vec[i] <- bf_cu
  bf_10_vec[i] <- bf_10
  bf_1c_vec[i] <- bf_1c
}

# Create data frame with results
results_df <- data.frame(
  Study = exList,
  bf_10 = bf_10_vec,
  bf_1u = bf_1u_vec,
  bf_1c = bf_1c_vec,
  bf_cu = bf_cu_vec
) %>%
  filter(Study != "Total")

# Create dataframe with only the relevant columns for the compute_joint_complements function
final_df <- results_df %>%
  select(Study, bf_1u, bf_cu) %>%
  mutate(P_theta_in_Hc = 0.5)

# T = nr of experiments
T <- nrow(final_df)
```

Study	$\mathrm{BF}_{1,0}^{(t)}$	$\mathrm{BF}_{1,u}^{(t)}$	$\mathrm{BF}_{1,c}^{(t)}$	$\mathrm{BF}_{c,u}^{(t)}$
Goldstein.2008.Ex1	1.453	1.941	32.670	0.059
Goldstein.2008.Ex2	2.023	1.974	75.336	0.026
Schulz2008.Ex2	0.354	1.720	6.148	0.280
Schulz2008.Ex3	0.580	1.858	13.045	0.142
Mair2010	0.316	1.225	1.579	0.775
Bohner2014.Ex1	0.065	0.667	0.501	1.333
Bohner2014.Ex2	0.064	0.036	0.018	1.964

Product Bayes factors

```
bf_10 <- prod(results_df$bf_10)
bf_1u <- prod(results_df$bf_1u)
bf_1c <- prod(results_df$bf_1c)
```

- $\mathrm{BF}_{1,0}^{(\mathrm{joint})} = 7.94e-04$.
- $\mathrm{BF}_{1,u}^{(\mathrm{joint})} = 0.3581$.
- $\mathrm{BF}_{1,c}^{(\mathrm{joint})} = 2.85e+03$.

BF against complete joint complement

- The complete complement consists of all $2^T - 1$ components H_Ω in which the prediction fails in at least one study.
- Its Bayes factor is a weighted average over these components (Eq. 7).
- Methods 1 and 2 both use equal weights and compute the same quantity in two ways. Method 3 uses complexity weights.

1: Equal weighting, enumeration (Eq. 7)

- Sums the Bayes factors of all $2^T - 1$ components, each with the same weight $\pi_\Omega = 1/(2^T - 1)$ (done inside `compute_joint_complements()`).
- Uses $\mathrm{BF}_{1,u}^{(t)}$ and $\mathrm{BF}_{c,u}^{(t)}$.
- Note: inside the function, the $\mathrm{BF}_{1,u}^{(t)}$ entering each component are recomputed from $\mathrm{BF}_{c,u}^{(t)}$ and $P(\theta^{(t)} \in H_c^{(t)})$ via Eq. 5. Here they equal the supplied $\mathrm{BF}_{1,u}^{(t)}$, because both come from the same posterior samples.

$$\mathrm{BF}_{c^*,u}^{(\mathrm{joint})} = \frac{1}{2^T - 1} \sum_{\emptyset \neq \Omega \subseteq \{1,\ldots,T\}} \prod_{t \in \Omega} \mathrm{BF}_{c,u}^{(t)} \prod_{t \notin \Omega} \mathrm{BF}_{1,u}^{(t)}, \qquad \mathrm{BF}_{1,c^*}^{(\mathrm{joint})} = \frac{\mathrm{BF}_{1,u}^{(\mathrm{joint})}}{\mathrm{BF}_{c^*,u}^{(\mathrm{joint})}} \quad \text{(Eq. 12)}$$

```
res <- compute_joint_complements(
  final_df,
  replace_0_with = 0.001,
  round_res = FALSE
)

# Sanity check: BF_1u(joint) / BF_cu(joint) should equal the product of the BF_1c's
all.equal(res["BES_c", "bf_1u"] / res["BES_c", "bf_cu"], bf_1c)
```

```
[1] TRUE
```

```
# BF_1c*(joint) = BF_1u(joint) / BF_c*u(joint) (transitivity, Eq. 12)
bf_1c_star_enumeration <- res["BES_c", "bf_1u"] / res["BES_c_star", "bf_cu_star"]
```

2: Equal weighting, simplified expression (Eq. 11)

- Same quantity as Method 1, algebraically simplified (Appendix A.2) so that the $2^T - 1$ components need not be enumerated.
- Uses only $\mathrm{BF}_{1,c}^{(t)} = \mathrm{BF}_{1,u}^{(t)} / \mathrm{BF}_{c,u}^{(t)}$: the same information as Method 1, reduced to one ratio per study.

$$\mathrm{BF}_{1,c^*}^{(\mathrm{joint})} = \frac{2^T - 1}{\prod_{t=1}^{T} (1 + 1/\mathrm{BF}_{1,c}^{(t)}) - 1}$$

```
bf_1c_star_formula_equal <- (2^T - 1) / (prod(1 + 1 / results_df$bf_1c) - 1)

# Check: identical to Method 1 (enumeration)
all.equal(bf_1c_star_formula_equal, bf_1c_star_enumeration)
```

```
[1] TRUE
```

3: Complexity weighting (Eq. 10)

- Weights each component by its share of the prior mass of the complete complement, $\pi_\Omega = c_\Omega / (1 - \prod_t c_1^{(t)})$.
- Uses $\mathrm{BF}_{1,u}^{(t)}$ and complexities $c_1^{(t)}$.
- Here $c_1^{(t)} = 0.5$ in all studies, so every component has the same prior mass 2^{-T} and complexity weighting reduces exactly to equal weighting.

$$\mathrm{BF}_{1,c^*}^{(\mathrm{joint})} = \frac{\prod_{t=1}^{T} \mathrm{BF}_{1,u}^{(t)} \, (1 - \prod_{t=1}^{T} c_1^{(t)})}{1 - \prod_{t=1}^{T} c_1^{(t)} \, \mathrm{BF}_{1,u}^{(t)}}$$

```
c1 <- 1 - final_df$P_theta_in_Hc

bf_1c_star_formula_complexity <-
  prod(results_df$bf_1u) * (1 - prod(c1)) / (1 - prod(c1 * results_df$bf_1u))

# Check: identical to equal weighting here, because c_1 = 0.5 in all studies
all.equal(bf_1c_star_formula_complexity, bf_1c_star_formula_equal)
```

```
[1] TRUE
```

Summary

$\mathrm{BF}_{1,0}^{(\mathrm{joint})}$	7.942e-04
$\mathrm{BF}_{1,u}^{(\mathrm{joint})}$	0.3581
$\mathrm{BF}_{1,c}^{(\mathrm{joint})}$	2.846e+03
$\mathrm{BF}_{1,c^*}^{(\mathrm{joint})}$, equal weighting, enumeration (Eq. 7)	0.3563
$\mathrm{BF}_{1,c^*}^{(\mathrm{joint})}$, equal weighting, simplified expression (Eq. 11)	0.3563

$\mathrm{BF}^{(\mathrm{joint})}_{1,c^*}$, complexity weighting (Eq. 10)	0.3563

Session info

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.5.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Berlin
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices datasets  utils     methods   base

other attached packages:
[1] stringr_1.6.0          tibble_3.3.1           here_1.0.2             dplyr_1.2.1
[5] BayesFactor_0.9.12-4.8 Matrix_1.7-5           coda_0.19-4.1

loaded via a namespace (and not attached):
 [1] jsonlite_2.0.0     compiler_4.6.1     renv_1.2.4         tidyselect_1.2.1   Rcpp_1.1.2
 [6] MatrixModels_0.5-4 parallel_4.6.1     jquerylib_0.1.4    yaml_2.3.12        fastmap_1.2.0
[11] lattice_0.22-9     R6_2.6.1           generics_0.1.4     knitr_1.52         rprojroot_2.1.1
[16] bslib_0.12.0       pillar_1.11.1      rlang_1.3.0        cachem_1.1.0       stringi_1.8.9
[21] xfun_0.61          sass_0.4.10        otel_0.2.0         cli_3.6.6          withr_3.0.3
[26] magrittr_2.0.5     digest_0.6.39      grid_4.6.1         pbapply_1.7-5      mvtnorm_1.4-2
[31] lifecycle_1.0.5    vctrs_0.7.3        evaluate_1.0.5     glue_1.8.1         rmarkdown_2.32
[36] tools_4.6.1        pkgconfig_2.0.3    htmltools_0.5.9
```